

Common Types of Cancer

Introduction

Cancer is not a single disease but a group of more than 100 diseases characterized by the uncontrolled growth and spread of abnormal cells. Different cancers originate in different organs or tissues of the body. Each type has unique symptoms, causes, diagnostic methods, treatment options, and survival rates. Early detection and timely treatment significantly improve patient outcomes and quality of life.

1. Breast Cancer

Overview

Breast cancer develops when abnormal cells in the breast grow uncontrollably, forming a tumor. Although it is most common in women, men can also develop breast cancer. It is one of the leading causes of cancer-related deaths among women worldwide.

Symptoms

- Lump or thickening in the breast or underarm
- Change in breast size or shape
- Skin dimpling or redness
- Nipple discharge (especially bloody)
- Inverted nipple
- Persistent breast pain

Risk Factors

- Female gender
- Increasing age
- Family history of breast cancer
- BRCA1 and BRCA2 gene mutations
- Obesity
- Alcohol consumption
- Hormone replacement therapy
- Late pregnancy or no pregnancy

Diagnosis

- Clinical breast examination
- Mammography
- Breast ultrasound

- Magnetic Resonance Imaging (MRI)
- Fine Needle Aspiration Cytology (FNAC)
- Core needle biopsy

Treatment

- Breast-conserving surgery (Lumpectomy)
- Mastectomy
- Radiation therapy
- Chemotherapy
- Hormone therapy
- Targeted therapy
- Immunotherapy

Prevention

- Regular breast self-examination
 - Mammography screening
 - Maintaining healthy body weight
 - Regular physical exercise
 - Limiting alcohol intake
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2. Lung Cancer

Overview

Lung cancer begins in the lungs and is one of the most common and deadliest cancers worldwide. Smoking remains the leading cause, although non-smokers can also develop lung cancer.

Symptoms

- Persistent cough
- Coughing up blood
- Chest pain
- Shortness of breath
- Wheezing
- Hoarseness
- Weight loss
- Fatigue

Risk Factors

- Cigarette smoking
- Passive smoking
- Air pollution
- Radon gas exposure
- Occupational exposure to asbestos
- Family history

Diagnosis

- Chest X-ray
- CT Scan
- PET Scan
- Bronchoscopy
- Sputum cytology
- Lung biopsy

Treatment

- Surgery
- Chemotherapy
- Radiation therapy
- Targeted therapy
- Immunotherapy

Prevention

- Avoid tobacco
- Reduce exposure to pollutants
- Use protective equipment in hazardous workplaces
- Regular health check-ups for high-risk individuals

3. Colorectal Cancer

Overview

Colorectal cancer develops in the colon or rectum and usually begins as small benign polyps that may become cancerous over time.

Symptoms

- Blood in stool
- Persistent diarrhea or constipation
- Abdominal discomfort
- Unexplained weight loss
- Fatigue
- Narrow stools

Risk Factors

- Age over 50 years
- Family history
- Low-fiber diet
- Obesity
- Smoking
- Alcohol consumption
- Physical inactivity

Diagnosis

- Colonoscopy
- Sigmoidoscopy
- Stool occult blood test
- CT Colonography
- Biopsy

Treatment

- Surgery
- Chemotherapy
- Radiation therapy
- Targeted therapy

Prevention

- High-fiber diet
- Regular exercise
- Colonoscopy screening
- Avoid smoking
- Limit alcohol consumption

4. Prostate Cancer

Overview

Prostate cancer occurs in the prostate gland of men and is one of the most common male cancers.

Symptoms

- Difficulty urinating
- Weak urine flow
- Frequent urination
- Blood in urine
- Blood in semen
- Pelvic discomfort
- Bone pain (advanced stages)

Risk Factors

- Increasing age
- Family history
- Obesity
- High-fat diet
- African ancestry

Diagnosis

- PSA Blood Test
- Digital Rectal Examination (DRE)
- MRI Scan
- Prostate biopsy

Treatment

- Active surveillance
- Surgery
- Radiation therapy
- Hormone therapy
- Chemotherapy

5. Cervical Cancer

Overview

Cervical cancer develops in the cervix, usually due to persistent infection with Human Papillomavirus (HPV).

Symptoms

- Abnormal vaginal bleeding
- Bleeding after intercourse
- Pelvic pain
- Vaginal discharge
- Pain during urination

Risk Factors

- HPV infection
- Smoking
- Multiple sexual partners
- Weak immune system

Diagnosis

- Pap Smear Test
- HPV DNA Test
- Colposcopy
- Cervical biopsy

Treatment

- Surgery
- Radiation therapy
- Chemotherapy
- Targeted therapy

Prevention

- HPV vaccination
- Regular Pap smear screening
- Safe sexual practices

6. Leukemia

Overview

Leukemia is a cancer of blood-forming tissues, including bone marrow and lymphatic system. It affects white blood cells, reducing the body's ability to fight infections.

Symptoms

- Frequent infections
- Fever
- Fatigue
- Easy bruising
- Bone pain
- Swollen lymph nodes
- Night sweats

Diagnosis

- Complete Blood Count (CBC)
- Bone marrow biopsy
- Flow cytometry
- Genetic testing

Treatment

- Chemotherapy
 - Targeted therapy
 - Immunotherapy
 - Stem cell transplantation
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7. Liver Cancer

Overview

Liver cancer starts in the liver cells and is commonly associated with chronic liver diseases.

Symptoms

- Jaundice
- Abdominal swelling
- Upper abdominal pain
- Weight loss
- Loss of appetite
- Fatigue

Risk Factors

- Hepatitis B infection
- Hepatitis C infection
- Liver cirrhosis
- Alcohol abuse
- Fatty liver disease

Diagnosis

- Ultrasound
- CT Scan
- MRI
- AFP Blood Test
- Liver biopsy

Treatment

- Surgery
 - Liver transplantation
 - Ablation
 - Chemotherapy
 - Targeted therapy
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8. Stomach Cancer**Overview**

Stomach cancer begins in the lining of the stomach and often develops slowly over many years.

Symptoms

- Persistent indigestion
- Stomach pain
- Nausea
- Vomiting
- Difficulty swallowing
- Weight loss

Risk Factors

- Helicobacter pylori infection

- Smoking
- High-salt diet
- Family history

Diagnosis

- Endoscopy
- Biopsy
- CT Scan
- PET Scan

Treatment

- Surgery
 - Chemotherapy
 - Radiation therapy
 - Targeted therapy
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9. Skin Cancer

Overview

Skin cancer develops when skin cells are damaged by ultraviolet (UV) radiation.

Symptoms

- New mole
- Changing mole
- Bleeding skin lesion
- Itching
- Non-healing sore

Risk Factors

- Excessive sun exposure
- Fair skin
- Family history
- Use of tanning beds

Diagnosis

- Skin examination
- Dermoscopy

- Skin biopsy

Treatment

- Surgical excision
- Cryotherapy
- Radiation therapy
- Immunotherapy

Prevention

- Use sunscreen
 - Wear protective clothing
 - Avoid prolonged sunlight exposure
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10. Thyroid Cancer

Overview

Thyroid cancer develops in the thyroid gland located at the base of the neck.

Symptoms

- Lump in the neck
- Difficulty swallowing
- Hoarseness
- Neck pain

Risk Factors

- Radiation exposure
- Family history
- Female gender

Diagnosis

- Thyroid ultrasound
- Fine Needle Aspiration (FNA)
- Thyroid function tests
- CT/MRI (if needed)

Treatment

- Thyroid surgery
- Radioactive iodine therapy

- Hormone replacement therapy
 - Targeted therapy
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General Prevention Tips

- Avoid tobacco products.
- Maintain a healthy body weight.
- Eat a balanced diet rich in fruits and vegetables.
- Exercise regularly.
- Limit alcohol intake.
- Protect skin from excessive sunlight.
- Get vaccinated against HPV and Hepatitis B.
- Attend regular cancer screening programs.
- Seek medical attention for persistent or unusual symptoms.